

FUTURE FOUNDATION

Structured Coaching Lesson Planner

Motion (Complete Chapter — Rest & Motion, Distance & Displacement, Speed, Velocity, Uniform & Non-Uniform Motion, Acceleration, Equations of Motion, Graphs, Free Fall, Circular Motion)

Physics — Motion

Subject	Physics
Chapter / Unit	Motion
Topic	Motion (Complete Chapter — Rest & Motion, Distance & Displacement, Speed, Velocity, Uniform & Non-Uniform Motion, Acceleration, Equations of Motion, Graphs, Free Fall, Circular Motion)
Target Level	CBSE Class 9
Batch Type	Foundation
Board Focus	CBSE
Teaching Sessions	15
Board Exam Q&A Sessions	3
Total Sessions	15 Teaching + 3 Board = 18 Total
Session Duration	40 minutes each
Total Programme Duration	720 minutes
Teaching Methodology	Structured Coaching — Explicit Instruction + Worksheet Integration + Board Exam Mastery

Pedagogy: Structured coaching with explicit instruction, systematic worksheet integration, shortcut techniques, and retrieval practice. Part A builds the concept; Part B converts the concept into board-exam marks.

Teaching Planner Overview

#	Session Focus	Part	Learning Outcome
1	Motion, Rest & Reference Frames	Teaching	Student defines rest, motion, reference frame; gives 3 examples of each; justifies why motion is relative
2	Distance and Displacement	Teaching	Student computes distance and displacement for straight-line and circular paths
3	Distance–Displacement	Teaching	Student handles

	in Circular & Curved Paths		circular-path D&D without confusion; recognizes 2R trap
4	Speed and Average Speed	Teaching	Student converts units fluently; solves multi-leg average-speed problems
5	Velocity and Average Velocity	Teaching	Student distinguishes speed from velocity; computes avg velocity for closed paths (correctly as zero)
6	Uniform and Non-Uniform Motion	Teaching	Student identifies uniform vs non-uniform motion from data, graphs, and descriptions
7	Acceleration — Concept and Sign	Teaching	Student computes acceleration with correct sign; interprets retardation
8	First Equation of Motion ($v = u + at$)	Teaching	Student derives and applies $v=u+at$ to 6+ problem types
9	Second Equation of Motion ($s = ut + \frac{1}{2}at^2$)	Teaching	Student derives and applies $s=ut+\frac{1}{2}at^2$ including from-rest cases
10	Third Equation of Motion ($v^2 = u^2 + 2as$)	Teaching	Student selects the right equation (time-absent problems → third eq)
11	Distance–Time and Velocity–Time Graphs	Teaching	Student reads both graph types confidently; computes s from area
12	Free Fall and Motion Under Gravity	Teaching	Student solves drop, throw-up, and throw-down problems
13	Uniform Circular Motion	Teaching	Student explains why UCM has acceleration despite constant speed
14	Multi-Phase & Hybrid Motion Problems	Teaching	Student breaks hybrid problems into phases and solves each
15	Complete Worksheet Walkthrough & Mastery Review	Teaching	Student has solutions and errors noted for every worksheet question
16	Board Exam: MCQ & Short-Answer Focus	Board Exam Q&A	Student attempts 1-2 mark board questions with correct marking-scheme awareness
17	Board Exam: Long-Answer & Case Study	Board Exam Q&A	Student writes structured long

	Focus		answers that earn full marks per marking scheme
18	Board Exam: Mock Test, Master Q&A Table & Exam Strategy	Board Exam Q&A	Student completes a timed mock and has final board strategy internalized

PART A — TEACHING SESSIONS

Session 1: Motion, Rest & Reference Frames

Session Number	1
Part	A — Teaching
Duration	40 minutes
Title	Motion, Rest & Reference Frames
Learning Objectives	• Define motion and rest using reference-frame language • Give 3 examples each of motion and rest • Justify why motion is relative — not absolute
Materials Needed	Blackboard, Worksheet Q14 (motion is relative), Two toy cars for demo
Worksheet Questions Covered	Q14
Shortcuts Introduced	'With respect to' — always ask this before saying rest or motion
Common Errors Addressed	• Saying 'X is at rest' without naming a frame
Homework	Read concept doc: Distance & Displacement section
Next Session Preview	Distance vs Displacement — the first scalar-vector fight
Pedagogy Note	Follows I do → We do → You do progression; opens with a hook.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Opening hook	Train-window demo: are YOU moving, or is the platform moving?	Close your eyes. Imagine you're in a train at a station. The train next to yours starts moving. For a second — who's moving? You or them? ...That confusion is the whole chapter in one moment.
5–18 min	Explicit teaching (I do)	Define motion, rest, reference frame; board examples of relative motion	Watch my board. Motion = change in position with respect to a frame. Rest = no change. Key phrase:

			'with respect to.' Without a frame, rest and motion are meaningless.
18–28 min	Guided practice (We do)	Classify 8 scenarios — walking, sleeping, statue, train, Earth	Next — the Earth. You're sitting still, right? But the Earth orbits the Sun at 30 km/s. Are you at rest or in motion? ...Depends on the frame. This is Q14 territory.
28–35 min	Independent practice (You do)	Students list 5 rest examples and 5 motion examples, then swap	Your turn. Five and five. Seven minutes. Then swap with your neighbour — if they disagree, argue it out using the frame rule.
35–38 min	Check for understanding	Exit ticket: 'Can you be at rest and in motion at the same time? Explain.'	One sentence on your exit slip. If your answer has the word 'frame' in it, you're golden.
38–40 min	Homework & preview	Read concept doc pp. Distance-Displacement section	Homework: read the Distance-Displacement pages tonight. Next session we get into scalars, vectors, and the first big exam topic.

Session 2: Distance and Displacement

Session Number	2
Part	A — Teaching
Duration	40 minutes
Title	Distance and Displacement
Learning Objectives	• Define distance and displacement precisely • Identify distance as scalar and displacement as vector • Compute both for straight-line paths with and without direction change
Materials Needed	Blackboard, Worksheet Q2, Q6, Q13, Q37, Ruler
Worksheet Questions Covered	Q2, Q6, Q13, Q37
Shortcuts Introduced	Has direction? → vector (displacement). No direction? → scalar (distance)., If back to start → displacement = 0, always.
Common Errors Addressed	• Confusing distance and displacement • Forgetting that displacement can be negative
Homework	Solve Q38 from worksheet
Next Session Preview	Distance-displacement in circles — the 2R trap
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	Books closed — 3 rapid questions on frame of reference	Books closed. What does 'motion is relative' mean? Give me one example. Define frame of reference. Go.
5–18 min	Explicit teaching (I do)	Define distance & displacement; board example of someone walking 3m east then 4m north	Watch. I walk 3 m east, then 4 m north. Distance = $3+4 = 7$ m. Displacement = $\sqrt{3^2+4^2} = 5$ m, northeast. Distance is the path — displacement is the arrow from start to end.
18–28 min	Guided practice (We do)	Work Q2 and Q6 on the board together	Open to Q2. Which is scalar? ...Distance. Why? ...No direction. Now Q6 — body returns to start. Displacement? ...Zero. Distance? ...Non-zero. Two different worlds.
28–35 min	Independent practice (You do)	Students solve Q13 and Q37 (assertion-reason)	Seven minutes. Q13 — which are vectors. Q37 — assertion about distance being scalar. Show reasoning beside each answer.
35–38 min	Check for understanding	Poll: 'If distance = displacement, what can you say about the path?'	Show of hands. Straight line, no turning back? ...Right. Distance = displacement only when path is straight AND no reversal.
38–40 min	Homework & preview	Assign Q38 (assertion on motion is relative)	Homework: Q38. Next session — the circular-path trap. Bring your brain.

Session 3: Distance–Displacement in Circular & Curved Paths

Session Number	3
Part	A — Teaching
Duration	40 minutes
Title	Distance–Displacement in Circular & Curved Paths
Learning Objectives	• Compute D and ΔD for quarter, semi, and full circles • Recognize the semicircle trap: displacement =

	$2R$, not πR • Solve multi-loop problems (n full loops $\rightarrow$ displacement = 0)
Materials Needed	Blackboard with compass, Worksheet Q73 (semicircular trip)
Worksheet Questions Covered	Q73
Shortcuts Introduced	Semicircle displacement = $2R$ (diameter) — NOT πR , n full loops $\rightarrow$ displacement = 0, distance = $n \times 2\pi R$
Common Errors Addressed	• Writing πR for semicircle displacement • Missing that full loops give zero displacement
Homework	Worksheet Q73
Next Session Preview	Speed — definition, formula, unit conversion
Pedagogy Note	Follows I do $\rightarrow$ We do $\rightarrow$ You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on scalar/vector, plus 'when is distance = displacement?'	Books closed. Scalar or vector: distance, displacement, velocity. When are distance and displacement equal? Go.
5–18 min	Explicit teaching (I do)	Draw semicircle of radius R . Walk through distance (πR) vs displacement ($2R$).	Watch my compass. Semi-circle, radius R . Distance — length of the curve = πR . Displacement — straight line from start to end = diameter = $2R$. Notice: distance > displacement, always on a curve.
18–28 min	Guided practice (We do)	Old-man-on-semicircle problem (radius 40 m); athlete running 5 full rounds	Old man, semicircle, $R=40$. Displacement = $2R = 80$ m. Now the athlete — 5 full rounds. Displacement? ...Zero. Distance? ... $2\pi R \times 5$. Two very different answers from similar wording.
28–35 min	Independent practice (You do)	Students solve quarter-circle problem and multi-loop problem	Your turn. Quarter circle radius R — what's distance? Displacement? Seven minutes. Draw it before you compute.
35–38 min	Check for understanding	Exit ticket: a cyclist does 2.5 laps on a circle of radius 10 m. Distance?	Exit slip. 2.5 laps, radius 10. Don't panic. Draw first.

		Displacement?	
38–40 min	Homework & preview	Worksheet Q73	Homework: Q73 — the 100 km round trip. Next session: speed arrives.

Session 4: Speed and Average Speed

Session Number	4
Part	A — Teaching
Duration	40 minutes
Title	Speed and Average Speed
Learning Objectives	• Define speed; state SI and CGS units • Convert between km/h and m/s fluently • Compute average speed for multi-leg journeys
Materials Needed	Blackboard, Worksheet Q8, Q25, Q26, Q27, Q29
Worksheet Questions Covered	Q8, Q25, Q26, Q27, Q29
Shortcuts Introduced	km/h → m/s: multiply by 5/18, m/s → km/h: multiply by 18/5
Common Errors Addressed	• Using 18/5 instead of 5/18 (wrong direction) • Forgetting to write formula line
Homework	Worksheet Q28
Next Session Preview	Velocity and average velocity — the vector cousin of speed
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 rapid questions on D&D including the circular trap	Books closed. Semicircle radius 10 — displacement? Distance? Now — can distance ever be less than displacement? ...Right, never.
5–18 min	Explicit teaching (I do)	Speed formula, units, km/h→m/s conversion	Watch. 1 km/h = 1000 m / 3600 s = 5/18 m/s. So 36 km/h = 36 × 5/18 = 10 m/s. Burn '5/18' into your head — it's half of every numerical this chapter.
18–28 min	Guided practice (We do)	Work Q25 (36 km/h → m/s) and Q8 (cyclist 100m/5s) together	Open to Q25. Convert. What factor? 5/18. Plug in. ...10 m/s. Now Q8 — cyclist. 100 m in 5 s. One step: divide. ...20 m/s.
28–35 min	Independent practice (You do)	Students solve Q26, Q27, Q29 under timer	Your turn. Three problems, seven

			minutes. Q26, Q27, Q29. Show the formula line first — that's a board-exam habit, start now.
35–38 min	Check for understanding	Exit ticket: 72 km/h = ? m/s	Exit slip. 72 km/h in m/s. One answer. Use the shortcut.
38–40 min	Homework & preview	Worksheet Q28	Homework: Q28. Next — velocity. Same formula, different personality.

Session 5: Velocity and Average Velocity

Session Number	5
Part	A — Teaching
Duration	40 minutes
Title	Velocity and Average Velocity
Learning Objectives	• Define velocity as vector; contrast with speed • Compute average velocity for round trips (often zero) • Interpret negative velocity correctly
Materials Needed	Blackboard, Worksheet Q16, Q39, Q73
Worksheet Questions Covered	Q16, Q39, Q73
Shortcuts Introduced	Avg velocity for round trip = 0 (displacement zero), Avg speed $\neq$ arithmetic average of two speeds for same-distance legs; use $2ab/(a+b)$
Common Errors Addressed	• Averaging speeds directly when distances are equal • Confusing 'avg velocity zero' with 'not moving'
Homework	Worksheet Q47
Next Session Preview	Uniform vs non-uniform motion — and graphs begin
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 rapid questions on speed and km/h conversion	Books closed. 54 km/h in m/s? 20 m/s in km/h? If speed is constant, what kind of motion? Go.
5–18 min	Explicit teaching (I do)	Velocity = displacement/time; sign matters; average velocity for closed loop	Watch. Velocity = displacement $\div$ time. Speed = distance $\div$ time. So for a round trip — speed non-zero, velocity ZERO. Because displacement is zero. Write that in bold.

18–28 min	Guided practice (We do)	Work Q16 (when is avg velocity zero?) and Q39	Q16 — when is avg velocity zero? ...Right, displacement zero. Body returns to start. Q39 — assertion-reason. R explains A? ...Yes, because displacement can be zero even with non-zero distance.
28–35 min	Independent practice (You do)	Q73: 100 km at 50 km/h + return at 60 km/h → avg speed & avg velocity	Q73 — big one. 100 km each way. Avg speed ≠ 55. Why? Time spent at slower speed is more. Seven minutes. Use total distance ÷ total time.
35–38 min	Check for understanding	Quick poll: 'Round trip — avg speed?' 'Round trip — avg velocity?'	Show of hands. Round trip, avg speed — non-zero, yes or no? Avg velocity — zero, yes or no? Both hands up on the right answer.
38–40 min	Homework & preview	Worksheet Q47	Homework: Q47. Next — uniform vs non-uniform. Graphs are coming.

Session 6: Uniform and Non-Uniform Motion

Session Number	6
Part	A — Teaching
Duration	40 minutes
Title	Uniform and Non-Uniform Motion
Learning Objectives	• Define uniform and non-uniform motion • Recognize uniform motion in tabulated data • Distinguish them using distance-time patterns
Materials Needed	Blackboard with tables, Worksheet Q3, Q15, Q18, Q40, Q63
Worksheet Questions Covered	Q3, Q15, Q18, Q40, Q63
Shortcuts Introduced	Uniform = 'equal distances in equal times', Non-uniform = at least one time interval breaks the pattern
Common Errors Addressed	• Calling 'constant speed on a curve' uniform motion (it's NOT — direction changes) • Confusing 'uniform' with 'straight-line'
Homework	Read concept doc: Acceleration section
Next Session Preview	Acceleration — the trickiest signed quantity in this chapter
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on speed vs velocity; when is avg velocity zero	Books closed. Speed vs velocity — one difference. Can avg velocity be zero while you're moving? Explain in one line. Go.
5–18 min	Explicit teaching (I do)	Tabular data: uniform = equal distance in equal time; non-uniform otherwise	Watch this table: 10 m in 1 s, 20 m in 2 s, 30 m in 3 s. Uniform — equal distance per equal time. Now: 5 m, 15 m, 30 m in 1s each. Non-uniform. The pattern breaks.
18–28 min	Guided practice (We do)	Work Q3, Q15, Q18 — uniform, non-uniform concept questions	Q3 — constant velocity means uniform. Q15 MCAT — which statements hold in uniform motion? Walk me through each option.
28–35 min	Independent practice (You do)	Students do Q40 (assertion-reason) and Q63 (matrix match)	Your turn. Q40 assertion-reason. Q63 matrix match — motion type vs characteristic. Seven minutes.
35–38 min	Check for understanding	Exit slip: 'Give one real-world example of uniform motion and one of non-uniform.'	One sentence for each. The best example of uniform motion isn't a car — think harder.
38–40 min	Homework & preview	Read acceleration section	Homework: read the acceleration section tonight. Next — the engine of this chapter: acceleration.

Session 7: Acceleration — Concept and Sign

Session Number	7
Part	A — Teaching
Duration	40 minutes
Title	Acceleration — Concept and Sign
Learning Objectives	• Define acceleration; state units and formula • Compute positive and negative acceleration correctly • Distinguish acceleration from retardation
Materials Needed	Blackboard, Worksheet Q1, Q7, Q20, Q30, Q43
Worksheet Questions Covered	Q1, Q7, Q20, Q30, Q43
Shortcuts Introduced	Brakes → negative a , always, Retardation = magnitude of negative acceleration (report

	positive value)
Common Errors Addressed	• Dropping the negative sign in deceleration problems • Writing m/s instead of m/s^2 as unit
Homework	Read concept doc: Equations of Motion section
Next Session Preview	First equation of motion — $v = u + at$
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on uniform motion	Books closed. Uniform motion — three properties. Difference from non-uniform. Give an example.
5–18 min	Explicit teaching (I do)	Acceleration = change in velocity ÷ time; units m/s^2 ; sign convention	Watch. $a = (v - u) / t$. If $v > u \rightarrow$ positive. If $v < u \rightarrow$ negative = deceleration. Brakes ALWAYS give negative a. Unit: m/s^2 . Not m/s, not m^2/s . Circle m/s^2 in your notes.
18–28 min	Guided practice (We do)	Q1 (SI unit), Q7 (brakes -5 m/s^2), Q20 (stone thrown up)	Q1 — SI unit. One word. Q7 — brakes from 20 m/s to 0 in 4 s. Sign of a? Negative. Value? -5 . Q20 — stone up. At peak, $v=0$. Acceleration? Still g, downward.
28–35 min	Independent practice (You do)	Q30 (stone up, total time 8s $\rightarrow$ peak at 4s) and Q43 (assertion on retardation)	Your turn. Q30, Q43. Seven minutes. Sign conventions — don't drop them.
35–38 min	Check for understanding	Exit ticket: a car slows from 30 m/s to 10 m/s in 5 s. Find a.	Exit slip. Answer with sign and units.
38–40 min	Homework & preview	Preview the three equations of motion	Homework: read about $v=u+at$ tonight. Next session we derive it and use it on six problems.

Session 8: First Equation of Motion ($v = u + at$)

Session Number	8
Part	A — Teaching
Duration	40 minutes
Title	First Equation of Motion ($v = u + at$)
Learning Objectives	• Derive $v = u + at$ from the definition of

	acceleration • Apply it to find v, u, a, or t in 6+ problem types • Identify problems where this is the right equation
Materials Needed	Blackboard, Worksheet Q9, Q10, Q33, Q43
Worksheet Questions Covered	Q9, Q10, Q33, Q43
Shortcuts Introduced	Time to stop = $u / a $
Common Errors Addressed	• Sign errors during rearrangement • Forgetting to verify with dimensional check
Homework	Derive $v = u + at$ in notebook
Next Session Preview	Second equation of motion — the displacement equation
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on acceleration: formula, unit, sign	Books closed. Acceleration formula. Unit. Sign of deceleration. Go.
5–18 min	Explicit teaching (I do)	Derive $v=u+at$ from $a=(v-u)/t$; demo Q9	Watch. $a = (v - u)/t$. Rearrange — multiply both sides by t, add u. Result: $v = u + at$. That's it. Now Q9: $u=10$, $a=2$, $t=3$. $v = 10 + 6 = 16$ m/s.
18–28 min	Guided practice (We do)	Q10 (train decelerating, find t) and Q33 (car stopping)	Q10 — train, $a=-0.2$, $u=20$, $v=0$. Find t. Rearrange: $t=(v-u)/a = -20/-0.2 = 100$ s. Sign cancellation — careful. Q33 — same setup.
28–35 min	Independent practice (You do)	Q43 (assertion-reason on $v=u+at$ usage)	Your turn. Q43 — assertion asks retardation. Use $v=u+at$ to verify. Seven minutes.
35–38 min	Check for understanding	Exit ticket: A bike accelerates from 5 m/s at 3 m/s^2 for 4 s. Find v.	Exit slip. One number with units.
38–40 min	Homework & preview	Preview second equation	Homework: derive $v=u+at$ yourself in your notebook. Next — displacement equation.

Session 9: Second Equation of Motion ($s = ut + \frac{1}{2}at^2$)

Session Number	9
Part	A — Teaching
Duration	40 minutes
Title	Second Equation of Motion ($s = ut + \frac{1}{2}at^2$)
Learning Objectives	• Derive $s = ut + \frac{1}{2}at^2$ graphically or algebraically • Apply to displacement problems including from-rest

	Recognize when to use this equation
Materials Needed	Blackboard, Worksheet Q21, Q32, Q45, Q48, Q76
Worksheet Questions Covered	Q21, Q32, Q45, Q48, Q76
Shortcuts Introduced	From rest ($u=0$): $s = \frac{1}{2}at^2$, Multi-phase → compute each phase separately, sum
Common Errors Addressed	- Forgetting the $\frac{1}{2}$ in $\frac{1}{2}at^2$ - Squaring t incorrectly
Homework	Read concept: $v^2 = u^2 + 2as$ section
Next Session Preview	Third equation — when time is the missing piece
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on $v=u+at$	Books closed. $v=u+at$. If $u=0$, what's v ? If $a=0$, what's v ? Car goes from 20 to 0 at -4 m/s^2 — how long?
5–18 min	Explicit teaching (I do)	Derive $s=ut+\frac{1}{2}at^2$; demo Q32 (from rest)	Watch. Displacement = avg velocity $\times$ time = $(u+v)/2 \times t$. Substitute $v=u+at$. Simplify. Result: $s = ut + \frac{1}{2}at^2$. Demo Q32 — from rest, so $u=0$. $s = \frac{1}{2} \times 2 \times 25 = 25 \text{ m}$.
18–28 min	Guided practice (We do)	Q21 (MCAT: $u=5$, $a=2$, $t=2$) and Q45 (assertion on $s=\frac{1}{2}at^2$)	Q21 — plug in $u=5$, $a=2$, $t=2$. $v=9 \checkmark$, $s=ut+\frac{1}{2}at^2=10+4=14 \checkmark$. Both (a) and (b). Q45 — assertion. From rest, $s=\frac{1}{2}at^2$. True, derived from full equation with $u=0$.
28–35 min	Independent practice (You do)	Q48 (assertion) and Q76 (multi-phase)	Q48 — assertion on $s=ut+\frac{1}{2}at^2$. Q76 — multi-phase: acceleration then constant. Split into phases. Seven minutes.
35–38 min	Check for understanding	Exit ticket: From rest, $a=4 \text{ m/s}^2$, $t=3 \text{ s}$. Find s .	Exit slip. Answer with units.
38–40 min	Homework & preview	Read third equation section	Homework: read $v^2=u^2+2as$ tonight. Next — the no-time-given equation.

Session 10: Third Equation of Motion ($v^2 = u^2 + 2as$)

Session Number	10
Part	A — Teaching
Duration	40 minutes
Title	Third Equation of Motion ($v^2 = u^2 + 2as$)
Learning Objectives	- Derive $v^2 = u^2 + 2as$ from the first two equations - Use it for no-time-given problems - Apply to stopping-distance problems
Materials Needed	Blackboard, Worksheet Q11, Q23, Q24, Q35, Q36, Q46
Worksheet Questions Covered	Q11, Q23, Q24, Q35, Q36, Q46
Shortcuts Introduced	No time? → use $v^2 = u^2 + 2as$, Stopping distance =

	$u^2 / (2 a)$
Common Errors Addressed	- Picking the wrong equation — forgetting to check if 't' is present - Sign errors on deceleration in $v^2=u^2+2as$
Homework	Worksheet Q70
Next Session Preview	Distance-time and velocity-time graphs
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on $s=ut+\frac{1}{2}at^2$	Books closed. $s=ut+\frac{1}{2}at^2$. What's s when $u=0$? When $a=0$? Write both.
5–18 min	Explicit teaching (I do)	Derive $v^2=u^2+2as$; demo Q35 (from rest, find v)	Watch. From $v=u+at$, solve for t. Sub into $s=ut+\frac{1}{2}at^2$. Simplify. Result: $v^2=u^2+2as$. Demo Q35 — $u=0$, $a=3$, $s=27$. $v^2=162$. $v\approx 12.73$ m/s.
18–28 min	Guided practice (We do)	Q11 (MCQ — which eq has no t), Q23 (MCAT eqs), Q36 (stopping dist)	Q11 — no-time eq. (c). Q23 — three eqs, all have 'a'. Q36 — bus stopping from 10 m/s, $a=-0.5$. Find s. Using $v^2=u^2+2as$ with $v=0$. $s = 100$ m.
28–35 min	Independent practice (You do)	Q24 (MCAT displacement forms), Q46 (assertion)	Your turn. Q24 — check each displacement formula. Q46 — assertion. Seven minutes.
35–38 min	Check for understanding	Exit ticket: car at 15 m/s brakes at -3 m/s ² . Find stopping distance.	Exit slip. No time given — which equation? ...Yes.
38–40 min	Homework & preview	Q70 (matrix match — equations vs scenarios)	Homework: Q70. Next — graphs. The picture version of everything we've learned.

Session 11: Distance–Time and Velocity–Time Graphs

Session Number	11
Part	A — Teaching
Duration	40 minutes
Title	Distance–Time and Velocity–Time Graphs
Learning Objectives	- Read d-t graphs: slope = speed; straight = uniform, curved = non-uniform - Read v-t graphs: slope = acceleration; area = displacement - Identify motion type from graph shape
Materials Needed	Blackboard with graph paper, Worksheet Q4, Q5, Q17, Q42, Q64
Worksheet Questions Covered	Q4, Q5, Q17, Q42, Q64
Shortcuts Introduced	d-t slope = speed, v-t slope = acceleration; v-t area

	= displacement
Common Errors Addressed	• Claiming d-t slope gives acceleration (WRONG — it gives speed) • Forgetting area under v-t represents displacement
Homework	Read concept: Free fall section
Next Session Preview	Free fall and motion under gravity
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on the three equations	Books closed. Write the three equations of motion. Go.
5–18 min	Explicit teaching (I do)	Draw d-t (uniform vs non-uniform) and v-t (accel) graphs; slope meaning	Watch my board. d-t graph. Straight line = uniform motion, slope = speed. Curved = non-uniform. Now v-t. Straight line = uniform acceleration, slope = a, area under curve = displacement. TWO graphs, TWO meanings of slope — don't confuse.
18–28 min	Guided practice (We do)	Q4 (d-t slope), Q5 (compare slopes), Q17 (d-t properties)	Q4 — d-t slope is? Speed. Q5 — Car B slope 3, Car A slope 2. Who's faster? B. Q17 — three correct options; watch the trap in (d).
28–35 min	Independent practice (You do)	Q42 (assertion) and Q64 (matrix match graphs)	Q42 — assertion: d-t slope = speed. True. Reason — speed is scalar. True but not the explanation. So (b). Q64 — match each graph property.
35–38 min	Check for understanding	Exit ticket: from a v-t graph triangle $u=0$, $v=10$, $t=5$ — find a and s	Exit slip. Slope = a. Area = s. One line each.
38–40 min	Homework & preview	Preview free fall	Homework: read about motion under gravity. Next — things go up, things come down.

Session 12: Free Fall and Motion Under Gravity

Session Number	12
Part	A — Teaching
Duration	40 minutes
Title	Free Fall and Motion Under Gravity
Learning Objectives	• Apply equations of motion with $a = g = 9.8$ (or 10) m/s^2 • Handle upward-throw sign convention • Use peak-time symmetry
Materials Needed	Blackboard, Worksheet Q30, Q34, Q75, Q78
Worksheet Questions Covered	Q30, Q34, Q75, Q78
Shortcuts Introduced	Drop time: $t = \sqrt{2h/g}$, Peak-time symmetry: throw-up and fall-down take equal time
Common Errors Addressed	• Using $g=+10$ when motion is upward (should be -10) • Forgetting that $v=0$ at peak
Homework	Read concept: Uniform Circular Motion section
Next Session Preview	Uniform circular motion — speed constant, velocity NOT
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on v-t graphs	Books closed. v-t graph slope gives? Area under v-t gives? What shape is uniform acceleration on v-t?
5–18 min	Explicit teaching (I do)	Free fall: $a=g$, $u=0$. Throw up: $a=-g$, $v=0$ at peak. Demo Q34.	Watch. Drop from rest — $u=0$, $a=g=9.8$ down, positive. Throw up — u positive, $a=-g$. At peak, $v=0$. Q34: $h=19.6$, $g=9.8$. $t=\sqrt{2h/g}=\sqrt{4}=2$ s. Memorize $t=\sqrt{2h/g}$ for dropping.
18–28 min	Guided practice (We do)	Q30 (peak time symmetry) and Q75 (ball up at 20 m/s, find time and height)	Q30 — total time 8 s. Peak at half = 4 s. Q75 — $u=20$, $a=-10$. Peak: $v=0$. $t=2$ s. Height: $h=u^2/(2g)=400/20=20$ m.
28–35 min	Independent practice (You do)	Q78 (drop from 45 m with $g=9.8$)	Your turn. Q78 — drop from 45 m. Find t and v on impact. Use $g=9.8$. Seven minutes.
35–38 min	Check for understanding	Exit ticket: stone dropped from 20 m, $g=10$. Find t and v on impact.	Exit slip. Two numbers.
38–40 min	Homework & preview	Preview circular motion	Homework: read uniform circular motion section. Next — the motion that keeps changing direction.

Session 13: Uniform Circular Motion

Session Number	13
Part	A — Teaching
Duration	40 minutes
Title	Uniform Circular Motion
Learning Objectives	• Explain why UCM has acceleration despite constant speed • Identify centripetal direction (toward centre) • Compute centripetal acceleration using v^2/r
Materials Needed	Blackboard with circle, Worksheet Q22, Q41, Q65, Q81
Worksheet Questions Covered	Q22, Q41, Q65, Q81
Shortcuts Introduced	Centripetal acceleration: $a = v^2/r$, direction toward centre, Constant speed $\neq$ zero acceleration when direction changes
Common Errors Addressed	• Claiming UCM has zero acceleration (WRONG — has centripetal) • Forgetting direction of centripetal acceleration
Homework	Review all three equations of motion; sketch one free-body situation for each
Next Session Preview	Multi-phase problems — chapter boss level
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on free fall	Books closed. Drop from 20 m, $g=10$ — find t . Peak of upward throw — velocity? Acceleration at peak?
5–18 min	Explicit teaching (I do)	UCM: speed constant, direction changes → velocity changes → acceleration	Watch my circle. Speed 10 m/s, constant. But here at point A — direction is east. At point B — direction is north. Velocity changed. Change in velocity ÷ time = acceleration. It's real. It points toward the centre — centripetal. $a = v^2/r$.
18–28 min	Guided practice (We do)	Q22 (MCAT), Q41 (assertion on circle acceleration), Q81 (centripetal a calc)	Q22 — check each option. Q41 — assertion: circle with constant speed has acceleration. TRUE. Reason: direction changes. TRUE, and R explains A. Q81 — $v=10$, $r=20$.

			$a=v^2/r=100/20=5$ m/s^2 .
28–35 min	Independent practice (You do)	Q65 (matrix match scenarios)	Q65 — match scenarios to outcomes. Circle at const speed → changing velocity. Work through all four. Seven minutes.
35–38 min	Check for understanding	Exit ticket: car circling at 20 m/s, $r=50$ m — find centripetal acceleration	Exit slip. $a = v^2/r$. One number, with direction.
38–40 min	Homework & preview	Preview multi-phase hybrid problems	Homework: none specific — review all equations. Next session: combined problems. The chapter's boss level.

Session 14: Multi-Phase & Hybrid Motion Problems

Session Number	14
Part	A — Teaching
Duration	40 minutes
Title	Multi-Phase & Hybrid Motion Problems
Learning Objectives	• Break hybrid problems into phases with clear u, v, a for each • Handle train-bridge, multi-phase, stop-and-go, vertical throws • Select the correct equation per phase
Materials Needed	Blackboard, Worksheet Q31, Q44, Q74, Q77, Q79, Q82, Q83, Q84
Worksheet Questions Covered	Q31, Q44, Q74, Q77, Q79, Q82, Q83, Q84
Shortcuts Introduced	Train crossing = train length + obstacle length, Multi-phase: end of phase 1 = start of phase 2
Common Errors Addressed	• Forgetting train's own length when crossing • Not carrying phase-1 velocity into phase 2
Homework	Review entire worksheet in preparation for Session 15 walkthrough
Next Session Preview	Complete worksheet walkthrough — every question analyzed
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	3 questions on centripetal acceleration and g	Books closed. Centripetal acceleration formula. Direction. Drop from 45 m with $g=10$ — find t .
5–15 min	Explicit teaching (I do)	Train-bridge: total	Watch. Train 50 m,

		length = train + bridge. Demo Q31.	bridge 250 m, speed 10 m/s. Total to cross = 300 m. $t = 300/10 = 30$ s. Not 25. Not 250/10. Add BOTH lengths. Q44 assertion says it — mark (a).
15–25 min	Guided practice (We do)	Q79 (multi-phase: decelerate then constant) and Q84 (accel then decelerate)	Q79 — phase 1: $u=15$, $a=-0.5$, $t=10 \rightarrow v=10$, $s=125$. Phase 2: $10 \times 5 = 50$. Total 175. Q84 — phase 1: $a=10$, $t=5 \rightarrow v=50$. Phase 2: decel from 50 at -5 . Time to stop = 10 s.
25–33 min	Independent practice (You do)	Q74 (train+platform), Q77 (stop-and-go), Q82 (deceleration + verify), Q83 (from rest derived)	Your turn. Four problems, eight minutes. Phase-split explicitly. I'll walk around.
33–38 min	Check for understanding	Exit ticket: car accelerates from rest at 2 m/s^2 for 5 s, then constant for 10 s. Total distance?	Exit slip. Two phases. Show both calculations.
38–40 min	Homework & preview	Review worksheet before Session 15 walkthrough	Homework: look over the worksheet tonight. Next session — every single question walked through. Come prepared to ask about any you're unsure of.

Session 15: Complete Worksheet Walkthrough & Mastery Review

Session Number	15
Part	A — Teaching
Duration	40 minutes
Title	Complete Worksheet Walkthrough & Mastery Review
Learning Objectives	• Review every worksheet question with solution approach and common error • Identify personal weak spots before board-exam sessions • Confirm readiness for board-exam practice
Materials Needed	Full worksheet (Q1–Q84), Blackboard, Error log sheet
Worksheet Questions Covered	Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q15, Q16, Q17, Q18, Q19, Q20, Q21, Q22, Q23, Q24, Q25, Q26, Q27, Q28, Q29, Q30, Q31, Q32, Q33, Q34, Q35, Q36, Q37, Q38, Q39, Q40, Q41, Q42, Q43, Q44, Q45, Q46, Q47, Q48, Q49, Q50, Q51, Q52, Q53, Q54, Q55, Q56, Q57,

	Q58, Q59, Q60, Q61, Q62, Q63, Q64, Q65, Q66, Q67, Q68, Q69, Q70, Q71, Q72, Q73, Q74, Q75, Q76, Q77, Q78, Q79, Q80, Q81, Q82, Q83, Q84
Shortcuts Introduced	Error-log method: circle missed questions, revisit weekly
Common Errors Addressed	• All 84 questions reviewed
Homework	Compile personal error log; re-solve 3 hardest questions
Next Session Preview	Board Exam Part B: MCQ & short-answer focus
Pedagogy Note	Follows I do → We do → You do progression; includes retrieval warm-up.

Minute-by-Minute Coaching Plan

Time	Phase	Activity	Teacher Script
0–5 min	Retrieval warm-up	Comprehensive 5-question recall spanning all 14 prior sessions	Books closed. Five quick ones. 1) Semicircle displacement formula. 2) km/h to m/s factor. 3) Stopping distance formula. 4) Free-fall drop time. 5) Centripetal acceleration. Go.
5–10 min	Concept recap	Board summary of 5 key formula groups + sign conventions	On the board — the whole chapter in 5 boxes. Equations of motion. Free fall. Graphs. Circular. Distance vs displacement. Burn these five into your head.
10–35 min	Complete worksheet discussion	Walk through the full 84-question table — approach, answer, common errors, shortcuts	Now we walk the whole worksheet. I'll move fast. If I skip something you're unclear on — raise your hand. For Q7, the trap is the negative sign. For Q25, the 5/18 shortcut. For Q73, harmonic mean. Eye on the table.
35–38 min	Mastery check	3 mixed questions across chapter — timed	Three minutes. Three questions. Any you miss — circle. We'll hunt them down in the board sessions.
38–40 min	Bridge to board sessions	Preview Part B	Next session — Part B begins. Previous years' board questions. Bring your pen, your watch, and your nerve.

Shortcut Techniques Reference

Shortcut	When to Use	Description	Example
km/h $\leftrightarrow$ m/s conversion	Any numerical mixing km/h and m/s units (very common in board exams)	km/h $\rightarrow$ m/s: multiply by 5/18. m/s $\rightarrow$ km/h: multiply by 18/5.	54 km/h = $54 \times 5/18 = 15$ m/s. 20 m/s = $20 \times 18/5 = 72$ km/h.
No-time-given $\rightarrow$ third equation	When question gives u, v, a or u, v, s but no t	Use $v^2 = u^2 + 2as$. Avoids finding t.	Car brakes from 20 m/s at -4 m/s^2 — stopping distance? $0 = 400 + 2(-4)s \rightarrow s = 50$ m. No t needed.
Semicircle trap: displacement = 2R	Any semicircular-path problem	Distance along semicircle = πR . Displacement (diameter) = 2R. NOT πR .	Walk semicircle of R = 10 m: distance = $10\pi \approx 31.4$ m; displacement = 20 m.
Free-fall drop time	Object dropped from height h with g	$t = \sqrt{2h/g}$. Memorize: for $g = 10$, $h = 5 \rightarrow t = 1$ s; $h = 20 \rightarrow t = 2$ s; $h = 45 \rightarrow t = 3$ s; $h = 80 \rightarrow t = 4$ s.	Drop from 45 m tower ($g=10$): $t = \sqrt{(90/10)} = \sqrt{9} = 3$ s.
Peak-time symmetry	Object thrown vertically up, returning to same height	Time to peak = half of total flight time. Peak velocity = 0.	Ball thrown up, returns in 8 s $\rightarrow$ peak at 4 s.
Stopping distance shortcut	Vehicle braking from u to 0	Stopping distance = $u^2 / (2 \times \text{deceleration})$. Derived from $v^2 = u^2 + 2as$ with $v=0$.	Car at 15 m/s, brakes at -3 m/s^2 . Stopping distance = $225/(2 \times 3) = 37.5$ m.
Harmonic mean for same-distance two-leg trips	Going and returning at different speeds over same distance	Avg speed = $2ab/(a+b)$, NOT $(a+b)/2$.	100 km at 50 km/h, return at 60 km/h. Avg = $2 \times 50 \times 60 / 110 \approx 54.5$ km/h (not 55).

Final Coaching Recommendation

This planner delivers 15 teaching sessions + 3 board-exam sessions = 18 total $\times$ 40 minutes = 720 minutes of structured coaching.

Teaching sessions (Part A, Sessions 1–15) build the concept through explicit instruction, guided practice, and independent practice. The final teaching session (Session 15) systematically reviews every worksheet question.

Board Exam sessions (Part B, Sessions 16–18) convert the concept into exam marks using previous-year questions, marking-scheme walkthroughs, timed practice, and live correction. The final session includes the Master Board Q&A Table covering 15+ previous-year questions.

Deliver sessions in order. Do not skip retrieval warm-ups — they are the spaced-repetition engine of this programme. Assign homework at the end of every session and check it at the start of the next.